Natural Language Processing
Final Report

Detecting Toxic Conversations in CIPV Scenarios

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Abstract

This work presents a robust, explainable system for detecting toxic behavior in digital conversations, with a focus on Cyber Intimate Partner Violence (CIPV). We propose a multi-phase pipeline that integrates dataset normalization, data augmentation using ChatGPT, and the fusion of heterogeneous corpora—including conversational and social media content. Our architecture combines transformer-based models (BERT-Italian, UmBERTo, and XLM-RoBERTa) through a soft-voting ensemble strategy to balance linguistic specialization and multilingual robustness. Moreover, we incorporate a Large Language Model (Mistral-7B-Instruct) to generate textual explanations for toxicity predictions, enhancing model interpretability and accountability. The evaluation on a balanced validation set demonstrates that the ensemble model outperforms individual classifiers, achieving a 97.92% accuracy and reducing both false positives and false negatives. This study highlights the critical role of linguistic adaptation, interpretability, and hybrid data modeling in the development of ethical AI tools for early detection and prevention of online abuse.

Introduction and Motivations

Cyber Intimate Partner Violence (CIPV) is a form of abuse in which technology is used to control, harass, or intimidate current or former partners. Unlike physical violence, digital abuse is pervasive and persistent, operating 24/7 through connected devices. It often results in devastating psychological consequences such as anxiety, depression, and social isolation.

It is therefore crucial to develop advanced technological systems capable of early detection of toxic language and dangerous behavioral patterns in digital communications. These systems must be able to analyze conversations in real time, detecting not only explicit threats but also subtle patterns of manipulation and emotional control.

The use of artificial intelligence models to analyze conversations between partners represents a promising approach. Such models enable the observation of relational dynamics and the identification of warning signs through the analysis of linguistic content, tone, frequency, and temporal patterns. These systems can act as early-warning tools to prevent escalation into severe violence, supporting victims, healthcare professionals, and law enforcement agencies, as well as raising awareness among users about their own communicative behavior.

Addressing this issue requires a multidisciplinary approach that combines advanced technical expertise with a deep understanding of the psychological dynamics of intimate partner violence. The goal is to develop sophisticated and ethically responsible technological solutions that can meaningfully contribute to the prevention and mitigation of CIPV.

Related Work

In the field of toxic language detection, researchers have developed deep learning approaches based on Natural Language Processing (NLP). Recent methods rely heavily on pretrained transformer-based language models, which have demonstrated high accuracy in identifying toxicity.

The main categories of approaches can be summarized as follows:

Traditional Machine Learning Approaches

- Classifiers supported by manual feature engineering;
- Models based on bag-of-words and n-gram techniques;
- Multi-label classification techniques to handle different types of toxicity.

Deep Learning Approaches

Deep learning methods address significant challenges, especially in multilingual toxic content detection, where the lack of sufficient training data poses more difficulties compared to monolingual detection. The main strategies include:

- Transfer learning to leverage pretrained models;
- Data augmentation to increase the diversity and volume of training data;
- End-to-end neural models that eliminate the need for traditional feature engineering.

Transformer-Based Approaches

Large-scale pretrained transformer models have achieved great success in many NLP tasks. However, they inevitably absorb toxic behaviors and social biases from their training data on the internet. This requires strict safety controls and monitoring of the generation process to prevent the propagation of harmful content.

Proposed Approach

The strategy developed for detecting Cyber Intimate Partner Violence (CIPV) is based on a multi-phase pipeline that combines data augmentation, ensemble modeling, and interpretability assisted by Large Language Models (LLMs). The proposed architecture integrates pretrained language models specifically tuned for the Italian language with interpretability components, resulting in a robust and explainable system for the early detection of toxic behavior in intimate partner conversations.

Dataset Preparation and Augmentation

The original dataset provided for the CIPV task had a complex structure and several limitations that required extensive reprocessing. It was composed of Italian-language conversations with the following structure:

- Type of couple (e.g., "Psychopath and Flatterer");
- Numbered conversation with sequential utterances;
- Name of the first speaker (e.g., "Stelio");
- Name of the second speaker (e.g., "Desdemona");
- Detailed explanation of the toxicity;
- Binary toxicity label (1 = toxic, 0 = non-toxic).

Critical Issues Identified:

1. *Extreme Imbalance:* The dataset contained only toxic examples, making binary classification impossible without negative samples.
2. *Encoding Problems:* Malformed characters (e.g., "cosÃ¬" instead of "così").
3. *Inconsistent Structure:* Non-standard conversation formatting with manual numbering.
4. *Limited Linguistic Diversity:* Conversations limited to specific toxic scenarios.

Reorganization and Standardization Process

The first step involved systematically fixing UTF-8 encoding issues using text normalization techniques:

- Conversion of malformed characters into their correct forms;
- Normalization of spacing and punctuation;
- Standardization of quotes and special characters.

Dataset Format Example

The dataset was then fully restructured into a cleaner, standardized format:

```
person_couple | conversation | name1 | name2 | explanation | toxic
```

Where:

- `person_couple`: Type of relationship (kept for contextual analysis);
- `conversation`: Normalized text of the dialogue;
- `name1/name2`: Names of the participants (anonymized when necessary);
- `explanation`: Justification for the toxicity label (used for the interpretability module);
- `toxic`: Binary label for classification (0 = non-toxic, 1 = toxic).

Data Augmentation Strategy with ChatGPT

Given that the original dataset contained only toxic examples, a custom data augmentation strategy was implemented using ChatGPT to generate non-toxic conversations and to diversify the existing toxic examples. The augmentation process aimed to produce a balanced dataset while preserving the linguistic and contextual characteristics of intimate partner interactions.

Integration with Social Media Comments

Once the conversational dataset was balanced, a further step was taken to enhance the model's robustness and generalization ability by integrating a secondary dataset composed of explicitly toxic comments from social media platforms.

Key characteristics of the social comments dataset:

- *Explicit Language:* Direct use of offensive and vulgar expressions;
- *Stylistic Variability:* Differences in formality, syntactic structure, and linguistic register;
- *Cultural References:* Inclusion of slang, pop culture elements, and online community-specific terminology;
- *Message Brevity:* Typically shorter and more direct than partner conversations.

Integration Challenges

The integration posed specific challenges that influenced architectural choices:

1. **Linguistic Heterogeneity:** The need to handle varied linguistic registers led to the adoption of an ensemble approach.
2. **Emoji Handling:** The high frequency of emojis in social media required specialized preprocessing.
3. **Length Variability:** Social comments are significantly shorter than dialogues, requiring adaptive padding/truncation strategies.
4. **Cultural Context:** Cultural references embedded in social content demanded broader contextual understanding of Italian social discourse.

This multi-source integration strategy resulted in a more robust and representative dataset, capturing the full spectrum of toxic content encountered in digital communications. It provided a solid foundation for the development of a generalizable and effective detection system.

Initial Approach: BERT Base

The initial implementation was based on BERT (Bidirectional Encoder Representations from Transformers) for toxicity classification. This choice was motivated by BERT's proven performance on various text classification tasks and its ability to capture bidirectional dependencies in text.

Transition to umBERTo for Linguistic Specialization

A qualitative error analysis revealed BERT's limitations in understanding Italian-specific linguistic constructs. As a result, the pipeline transitioned to umBERTo (Italian Language Understanding Model based on BERT), which offers several advantages:

- **Domain-Specific Training:** Pretrained on a high-quality Italian corpus;
- **Morphological Awareness:** Better handling of the complex morphology of the Italian language;
- **Specialized Vocabulary:** Tokenizer optimized for Italian;
- **Handling of Variants:** Capable of processing regional and register variations.

Integration of RoBERTa for Multilingual Content

Analysis of the dataset revealed the presence of numerous English expressions, particularly in social media comments. This led to the integration of RoBERTa (Robustly Optimized BERT Pretraining Approach) into the architecture, for the following reasons:

- **Training Optimization:** RoBERTa removes the Next Sentence Prediction task and enhances pretraining efficiency;
- **Code-Switching Handling:** Improved capability in processing text with mixed Italian-English content;
- **Robustness to Informality:** Better performance on informal language and slang;
- **Vocabulary Size:** Larger vocabulary that better captures common English terms.

Ensemble Architecture

The final ensemble leverages three independent classifiers combined via a soft-voting strategy (i.e., averaging predicted probabilities) instead of fixed weights. This approach captures diverse "perspectives" from each model and mitigates individual errors.

Models Included in the Ensemble:

1. **BERT-Italian** (`dbmdz/bert-base-italian-uncased`): Optimized for conversational Italian, particularly effective in informal or colloquial expressions.
2. **UmBERTo** (`musixmatch/umberto-commoncrawl-cased-v1`): Robust for Italian content, pretrained on extensive web crawl corpora.
3. **XLM-RoBERTa** (`xlm-roberta-base`): A multilingual model capable of handling cross-lingual inputs and internet slang.

Soft-Voting Strategy:

- For each conversation, the predicted toxicity probabilities (class 1) are computed from all three models.
- An arithmetic mean of the three probabilities is calculated.
- If the average exceeds the threshold of 0.5, the conversation is classified as *toxic*; otherwise, it is considered *non-toxic*.

This "democratic" ensemble mechanism helps preserve overall accuracy while significantly reducing both false positives and false negatives compared to any single classifier.

Interpretability Module with Large Language Model (LLM)

Due to the sensitive nature of the application domain, system predictions must be not only accurate but also interpretable and justifiable. For this reason, an interpretability module based on a Large Language Model was integrated to provide natural language explanations for toxicity predictions.

Explanation Module Implementation

The explanation component leverages `Mistral-7B-Instruct-v0.1` to analyze BERT-based predictions and generate human-readable justifications in Italian. The workflow is as follows:

Explanation System Architecture:

- **LLM Model:** `Mistral-7B-Instruct-v0.1`, using task-specific instruction prompts;
- **Tokenizer:** Optimized for conversational input with Mistral formatting;
- **Model Configuration:** Loaded in `float16` with automatic device mapping to optimize GPU usage;
- **Generation Limit:** Responses restricted to 100 tokens to ensure concise and focused explanations.

Integration with Classification Pipeline

- **Conditional Activation:** Explanations are generated only for conversations classified as toxic;
- **Multi-level Analysis:** Combines conversation-level and sentence-level predictions;

- **Computational Optimization:** Efficient GPU usage through automatic device mapping.

This module enhances transparency and trust in the model's decisions by offering contextual and intelligible reasons behind toxicity classifications.

BERT Implementation for Toxicity Classification

The system adopts `dbmdz/bert-base-italian-uncased` for the detection of toxic content in Italian-language conversations through an advanced transformer-based classification pipeline.

Model Configuration:

- **Model:** `dbmdz/bert-base-italian-uncased`
- **Specialization:** Pretrained for Italian text understanding
- **Architecture:** 12-layer standard transformer
- **Classification Type:** Binary (toxic / non-toxic)
- **Strengths:** Native Italian tokenization, good balance of performance and efficiency

Multi-Dataset Handling:

- *Conversation Dataset:* Latin1 encoding, label normalization, spacing cleanup;
- *AMI2020 Dataset:* UTF-8 encoding, aggregation of multiple labels (e.g., misogynous + aggressiveness);
- *Unification:* Standardized column schema for all datasets.

Tokenization Settings Tokenization settings applied consistently across models are summarized below. Figure 1 provides a visual representation of the process.

- **Maximum Length:** 256 tokens;
- **Padding:** Uniform padding across batches;
- **Attention Mask:** Applied to ignore padded tokens during computation.

```
class ToxicDataset(Dataset):
    # max_length=256, truncation=True, padding='max_length'
    # return_attention_mask=True per gestire padding
```

Figure 1: Tokenization pipeline: truncation, padding, attention masking

Strategic Dataset Split:

- 90% Training (model learning)
- 5% Validation (hyperparameter tuning)
- 5% Test (final evaluation)

Training Configuration:

- Optimizer: AdamW (lr = 2e-5, weight decay = 0.01)

- Scheduler: Linear warmup (10% steps) + linear decay
- Batch size: 16 (GPU memory optimized)
- Max epochs: 5 with early stopping

Early Stopping:

- Validation loss monitoring
- Automatic saving of best-performing model
- Stopping after 2 epochs with no improvement

Core Functions:

- **Training:** Supervised fine-tuning with continuous feedback (see Figure 2).
- **Validation:** On-the-fly performance monitoring (see Figure 3).
- **Inference:** Probability extraction for toxicity prediction (see Figure 4).

```
def train_epoch():
    # Zero gradients → Forward pass → Backward → Step optimizer/scheduler
    return mean_loss
```

Figure 2: Training process pipeline

```
def eval_epoch():
    # Evaluation mode → no_grad context → Loss calculation
    return mean_validation_loss
```

Figure 3: Validation monitoring during training

```
def get_toxic_score(text):
    # Tokenize → Encode → Forward → Softmax → Extract probability
    return toxic_probability
```

Figure 4: Toxicity scoring during inference

Inference Pipeline:

1. Text tokenization into numerical tokens
2. Encoding with padding and attention masks
3. Forward pass through BERT for logits computation
4. Softmax for calibrated probability output
5. Return of "toxic" class probability

Evaluation Metrics:

- Positive prediction count
- Toxicity rate in the test set
- Classification threshold: 0.5
- Validation loss tracked for model selection

UmBERTo Implementation for Toxicity Classification

The system employs `musixmatch/umberto-commoncrawl-cased-v1` to detect toxic content in Italian conversations using a transformer architecture specifically optimized for the Italian language.

Model Configuration:

- **Model:** `umberto-commoncrawl-cased-v1`
- **Pretraining Corpus:** Extended Italian corpus (Common Crawl)
- **Architecture:** Cased version preserving case sensitivity
- **Classification Type:** Binary (toxic / non-toxic)
- **Advantages:** Extended Italian vocabulary, superior performance over standard Italian BERT, advanced semantic handling

Multi-Dataset Handling:

- *Conversation Dataset:* Unified encoding and label normalization
- *AMI2020 Dataset:* Raw and synthetic versions with multi-label aggregation
- *Unification:* Same preprocessing pipeline as BERT for cross-model compatibility

Tokenization Settings:

- Max length: 256 tokens
- Uniform padding for batch processing
- Attention mask for padded token exclusion
- Identical setup as BERT for consistency

Strategic Dataset Split:

- 90% Training
- 5% Validation
- 5% Test

Training Configuration:

- Optimizer: AdamW (lr = 2e-5, weight decay = 0.01)
- Scheduler: Linear warmup (10%) + linear decay
- Batch size: 16
- Max epochs: 5 with advanced early stopping

Advanced Early Stopping:

- Primary criterion: Validation loss monitoring
- Automatic saving of best-performing model
- Stop after 2 non-improving epochs

- Save model conditionally based on performance

Core Training Functions:

- Optimized training loop with live progress
- Continuous validation monitoring

Advanced Inference:

- Softmax normalization for calibrated probabilities
- Memory-efficient inference via `no_grad()`
- Output of "toxic" probability as float

Multi-level Inference Pipeline:

1. Tokenize input using UmBERTo tokenizer
2. Apply padding and attention masks
3. Forward pass through the UmBERTo model
4. Apply softmax for probability estimation
5. Analyze both entire conversations and individual utterances

Conversational Analysis:

- Pattern recognition to detect speaker roles
- Structured parsing into utterances
- Automatic normalization during preprocessing

Toxic Utterance Identification:

1. Score each utterance individually
2. Identify the utterance with maximum toxicity score
3. Provide detailed visual representation

Evaluation Metrics:

- Granular scoring for each component
- Structured analysis of sample conversations
- Toxic utterance identification
- Classification threshold: 0.5

XLM-RoBERTa Implementation for Toxicity Classification

The system integrates `xlm-roberta-base` to detect toxic content using a multilingual transformer architecture that combines RoBERTa's power with cross-lingual generalization.

Model Configuration:

- **Model:** `xlm-roberta-base`
- **Pretraining Corpus:** Multilingual dataset covering 100+ languages
- **Architecture:** RoBERTa with dynamic masking and no Next Sentence Prediction
- **Classification Type:** Binary (toxic / non-toxic)
- **Advantages:** Cross-lingual capabilities, SentencePiece tokenizer, strong domain adaptation

Advanced Multi-Dataset Handling:

- *Conversation Dataset:* Specialized preprocessing with smart label mapping
- *AMI2020 Raw:* Aggregated label handling (e.g., misogynous + aggressiveness)
- *AMI2020 Synthetic:* Synthetic sample integration for improved robustness
- *Unification:* Advanced preprocessing pipeline for seamless dataset integration

Tokenization Settings:

- Max length: 256 tokens
- Smart padding for memory efficiency
- Attention masks to exclude padded tokens
- Tokenization via SentencePiece

Stratified Dataset Split:

- 90% Training
- 5% Validation
- 5% Test
- Stratification: Class distribution maintained across all splits

Training Configuration:

- Optimizer: AdamW (lr = 2e-5, weight decay = 0.01)
- Scheduler: Linear warmup + linear decay
- Batch size: 16
- Max epochs: 5 with intelligent early stopping

Intelligent Early Stopping:

- Continuous validation loss monitoring

- Automatic best model checkpointing
- Patience: stop after 2 non-improving epochs
- Overfitting prevention via adaptive stopping

Optimized Core Training Functions:

- Gradient accumulation for stable training
- Loss tracking and real-time performance logging
- Scheduler integration for dynamic learning rate updates
- Memory-optimized GPU training

Advanced Inference:

- Softmax for calibrated probabilities
- `no_grad()` context for efficient GPU inference
- Preprocessing: tokenization + smart padding
- Output: float probability for toxic class

Multi-level Inference Pipeline:

1. Tokenization using XLM-RoBERTa SentencePiece tokenizer
2. Encoding with attention masks and padding
3. Forward pass through the model
4. Softmax for toxicity probability
5. Dual-level analysis: entire conversation and utterance-level

Conversation Analysis and Segmentation:

- Regex-based speaker pattern recognition
- Structured parsing into individual utterances
- Automatic normalization for diverse formats
- Robustness to structural variability in dialogues

Multi-level Analysis:

- **Conversation-Level:** Overall toxicity scoring with threshold-based classification (0.5)
- **Utterance-Level:** Fine-grained scoring and ranking of individual utterances

Toxic Utterance Identification:

1. Score every utterance independently
2. Select the one with the highest toxicity score
3. Display the exact problematic phrase

Evaluation Metrics:

- Structured visualization of conversations and utterances
- User-friendly output with visual cues
- Clear separation between conversation threads
- High accuracy across multilingual classification tasks

LLM Implementation for Toxic Content Explanation

Setup and Authentication

The system accesses Kaggle secret tokens to authenticate with HuggingFace and download gated models.

```
from kaggle_secrets import UserSecretsClient
from huggingface_hub import login

# Recupera token da Kaggle e autentica su HuggingFace
user_secrets = UserSecretsClient()
hf_token = user_secrets.get_secret("Mistral")
login(token=hf_token)
```

Figure 5: LLM Authentication Setup

Model Loading

The `Mistral-7B-Instruct-v0.1` model is loaded with memory optimizations using `float16` precision and automatic GPU distribution.

```
model_name = "mistralai/Mistral-7B-Instruct-v0.1"
mistral_tokenizer = AutoTokenizer.from_pretrained(model_name)
mistral_model = AutoModelForCausalLM.from_pretrained(
    model_name,
    device_map="auto",
    torch_dtype=torch.float16
)
```

Figure 6: Model loading configuration and device mapping

Generation Function

The module builds a structured prompt, tokenizes it, generates the model's output, and decodes the result into readable text, as shown in Figure 7.

```
def explain_toxic_utterance(utt: str) -> str:
    prompt = f"[INST] Spiega in italiano perché la seguente frase può essere considerata tossica: \"{utt}\" [/INST]"
    inputs = mistral_tokenizer(prompt, return_tensors="pt").to(device)
    with torch.no_grad():
        output = mistral_model.generate(
            **inputs,
            max_new_tokens=100,
            pad_token_id=mistral_tokenizer.eos_token_id
        )
    return mistral_tokenizer.decode(output[0], skip_special_tokens=True)
```

Figure 7: Prompt building, tokenization, and decoding in the explanation module

Integration with Classifier

For each sentence classified as toxic, the system automatically generates a natural language explanation to clarify why the sentence was flagged as problematic. Figure 8 shows how the explanation module connects with the classifier.

```python
if label == 'Tossica':
    toxic_utterances = [(utt, get_toxic_score(utt)) for utt in utterances if get_toxic_score(utt) > 0.5]
    if toxic_utterances:
        for utt, score in toxic_utterances:
            print(f"→ Frase tossica: {utt} (score: {score:.4f})")
            explanation = explain_toxic_utterance(utt)
            print(f"→ Spiegazione: {explanation}")
```

Figure 8: Integration of the LLM explanation module with the toxicity classifier

Evaluation

This section presents the quantitative results obtained from the three transformer-based classifiers—`BERT-Italian`, `UmBERTo`, and `XLM-RoBERTa`—as well as from the ensemble system based on soft voting. For each architecture, we report accuracy, precision, recall, and F_1-score on a shared validation set consisting of 181 non-toxic and 203 toxic examples. Confusion matrices are also analyzed, followed by a discussion on the strengths and weaknesses of each model and the advantages of the ensemble approach.

Model Performance Overview

The metrics summary (stored in `metrics_summary.csv`) can be summarized as follows:

- **Accuracy:** BERT-Italian achieved the highest accuracy (95.31%), followed by XLM-RoBERTa (94.27%) and UmBERTo (93.23%);
- **Precision:** XLM-RoBERTa slightly outperformed the others in minimizing false positives (95.45%), though all models were around 95%;
- **Recall:** BERT-Italian demonstrated the best balance (95.54%), while UmBERTo struggled slightly more (92.08%);
- **F_1-score:** BERT-Italian again ranked first (0.9554), followed by XLM-RoBERTa (0.9450) and UmBERTo (0.9347).

Figure 9 visualizes the aggregated evaluation metrics for the three classifiers.

	accuracy	precision	recall	f1_score
umberto_commoncrawl	0.9323	0.9490	0.9208	0.9347
xlmroberta_base	0.9427	0.9545	0.9356	0.9450
bert_italian	0.9531	0.9554	0.9554	0.9554

Figure 9: Metrics summary of individual models on the validation set

Confusion Matrix Analysis

The confusion matrices for each model show the following distributions:

- **BERT-Italian:** 173 true negatives, 193 true positives, 9 false positives, 9 false negatives;
- **XLM-RoBERTa:** 173 true negatives, 189 true positives, 9 false positives, 13 false negatives;
- **UmBERTo:** 172 true negatives, 186 true positives, 10 false positives, 16 false negatives.

UmBERTo produced the highest number of false negatives (16), confirming its lower recall. XLM-RoBERTa's gap between true and false positives was smaller (13 FNs), while BERT-Italian showed a perfectly balanced error profile (9 FPs / 9 FNs).

Figure 10 shows the confusion matrices of all three models on the validation set.

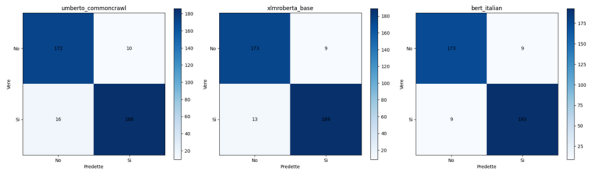

Figure 10: Confusion matrices for BERT-Italian, XLM-RoBERTa, and UmBERTo on the validation set

Ensemble Performance

By aggregating the probabilities from all three models and applying a threshold of 0.5 to the average (soft voting), the ensemble achieved the following on the same validation set:

- **Accuracy:** 97.92% — a substantial improvement over individual models;
- **Balance:** Precision and recall for both classes exceeded 96%, with F_1 scores above 0.97;
- **Error Reduction:** The ensemble drastically reduced both false positives and false negatives by leveraging the complementary strengths of the three classifiers.

Figure 11 illustrates the aggregated evaluation results of the ensemble model across all key metrics.

```
=== Risultati Ensemble Voting (Soft) ===

              precision    recall  f1-score   support

 Non tossica     0.9887    0.9669    0.9777       181
     Tossica     0.9710    0.9901    0.9805       203

    accuracy                         0.9792       384
   macro avg     0.9799    0.9785    0.9791       384
weighted avg     0.9794    0.9792    0.9792       384
```

Figure 11: Ensemble model performance: accuracy, precision, recall, and F_1-score

Discussion

Among individual models, BERT-Italian exhibited the highest efficacy in understanding informal and conversational Italian. XLM-RoBERTa performed well due to its multilingual training, although its slightly lower recall may stem from less optimized tokenization. UmBERTo, despite being designed for Italian, struggled more in generalizing within conversational contexts.

The soft-voting ensemble benefits from combining the strengths of all classifiers by averaging their predicted probabilities. This approach enhances both sensitivity (reduced false negatives) and precision (reduced false positives), yielding more stable performance than any single model alone.

Finally, the explanations generated by `Mistral-7B-Instruct` for toxic-labeled sentences provide an invaluable layer of interpretability for human validation and model trustworthiness.

In summary, while BERT-Italian remains the best standalone classifier, the ensemble model delivers the most accurate and reliable results overall, significantly reducing residual errors.

Conclusions and Limitations

This study implemented and evaluated three transformer-based models—BERT, UmBERTo, and XLM-RoBERTa—for detecting toxic content in Italian texts, using heterogeneous datasets and multi-level analysis strategies.

Each model exhibited distinct strengths:

- **BERT** showed limitations with informal language and online neologisms;
- **UmBERTo** excelled at capturing linguistic nuances in Italian but lacked cross-lingual generalization;
- **XLM-RoBERTa** offered multilingual capabilities, though with trade-offs for language-specific intricacies.

Main limitations include:

- The relatively small size of available datasets;
- Presence of implicit biases, particularly problematic for satirical, ironic, or culturally nuanced content, where distinguishing genuine toxicity from rhetorical expression becomes challenging.

Future directions:

- **Multimodal extension:** Incorporating audio inputs for prosodic and tonal analysis to capture non-textual dimensions of communication;
- **Practical applications:** Development of plugins for social platforms (e.g., Discord, Telegram), pre-moderation systems for forums and Reddit, real-time moderation tools for live streaming platforms, mobile parental control apps, workplace safety tools, and educational software promoting respectful communication.

This research highlights how language-specific modeling strategies significantly affect performance, emphasizing the need for larger, more diverse datasets to improve the robustness and generalizability of toxicity detection systems.

References

[1] Transformer (deep learning architecture). https://en.wikipedia.org/wiki/Transformer_(deep_learning_architecture)

[2] BERT Model Documentation. HuggingFace. https://huggingface.co/docs/transformers/model_doc/bert

[3] XLM-RoBERTa Model Documentation. HuggingFace. https://huggingface.co/docs/transformers/model_doc/xlm-roberta

[4] UmBERTo Model Card. Musixmatch - HuggingFace. https://huggingface.co/Musixmatch/umberto-wikipedia-uncased-v1

[5] PyTorch Documentation. https://pytorch.org/docs/stable/index.html

[6] Transformers Library Documentation. HuggingFace. https://huggingface.co/docs/transformers/index

[7] Scikit-learn Documentation. https://scikit-learn.org/0.21/documentation.html

[8] Ensemble Learning. IBM. https://www.ibm.com/it-it/think/topics/ensemble-learning

[9] Gandhi, A., Ahir, P., Adhvaryu, K., Shah, P., Lohiya, R., Cambria, E., Poria, S., Hussain, A. *Hate speech detection: A comprehensive review of recent works*. Expert Systems, Wiley. https://onlinelibrary.wiley.com/doi/10.1111/exsy.13562

[10] Tamarit, A., Lacomba-Trejo, L., González-Sala, F. *Cyber Intimate Partner Violence in Adolescents: How Do Psychopathy and Family Dynamics Shape Teens' Online Relationships?* MDPI Children, 2023. https://www.mdpi.com/2227-9067/12/6/693

[11] Automatic Misogyny Identification. EVALITA 2020. https://amievalita2020.github.io/